# Experiment 4 : Data Wrangling on Real Estate Marke

**Aim:** To perform data wrangling on a real estate dataset: clean column names, handle missing v and outliers, and aggregate housing prices by neighborhood and property type.

In [1]:
```
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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
np.random.seed(3)
n = 250
df = pd.DataFrame({
    "Property ID": range(1, n + 1),
    "Neighborhood ": np.random.choice(["Aundh", "Kothrud", "Wakad", "Hinj
    "Property-Type": np.random.choice(["Flat", "Villa", "Plot"], n),
    "Area (sqft)": np.random.randint(500, 3000, n),
    " Bedrooms": np.random.randint(1, 5, n),
    "Sale Price (Rs)": np.random.randint(3000000, 20000000, n),
    "Year Built": np.random.randint(1990, 2024, n),
})
df.head()
```

Out[2]:

| | Property ID | Neighborhood | Property-Type | Area (sqft) | Bedrooms | Sale Price (Rs) | Ye |
|---|---|---|---|---|---|---|---|
| 0 | 1 | Wakad | Villa | 604 | 1 | 10898035 | 2 |
| 1 | 2 | Aundh | Villa | 2153 | 3 | 16490355 | 2 |
| 2 | 3 | Kothrud | Villa | 2912 | 3 | 16767181 | 2 |
| 3 | 4 | Hinjewadi | Flat | 1239 | 2 | 85909209 | 1 |
| 4 | 5 | Aundh | Flat | 2065 | 3 | 17276029 | 2 |

In [3]:
```
df.loc[np.random.choice(n, 12), "Area (sqft)"] = np.nan
df.loc[np.random.choice(n, 4), "Sale Price (Rs)"] = 90000000
print("shape after adding missing values and outliers:", df.shape)
```

```
shape after adding missing values and outliers: (250, 7)
```

In [4]:
```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_").str
print(list(df.columns))
```

```
['property_id', 'neighborhood', 'property_type', 'area_sqft', 'bedrooms',
'sale_price_rs', 'year_built']
```

In [5]:
```
print("missing values:")
df.isnull().sum()
```

```
missing values:
```

Out[5]:
```
property_id       0
neighborhood      0
property_type     0
area_sqft        12
bedrooms          0
sale_price_rs     0
year_built        0
dtype: int64
```

In [6]:
```
df["area_sqft"] = df["area_sqft"].fillna(df["area_sqft"].median())
print("missing area_sqft now:", df["area_sqft"].isnull().sum())
```

```
missing area_sqft now: 0
```

In [7]:
```
q1 = df["sale_price_rs"].quantile(0.25)
q3 = df["sale_price_rs"].quantile(0.75)
iqr = q3 - q1
df = df[df["sale_price_rs"] <= q3 + 1.5 * iqr]
print("after outlier removal:", df.shape)
```

```
after outlier removal: (246, 7)
```

In [8]:
```
flats = df[df["property_type"] == "Flat"]
print("flats only:", flats.shape)
```

```
flats only: (81, 7)
```

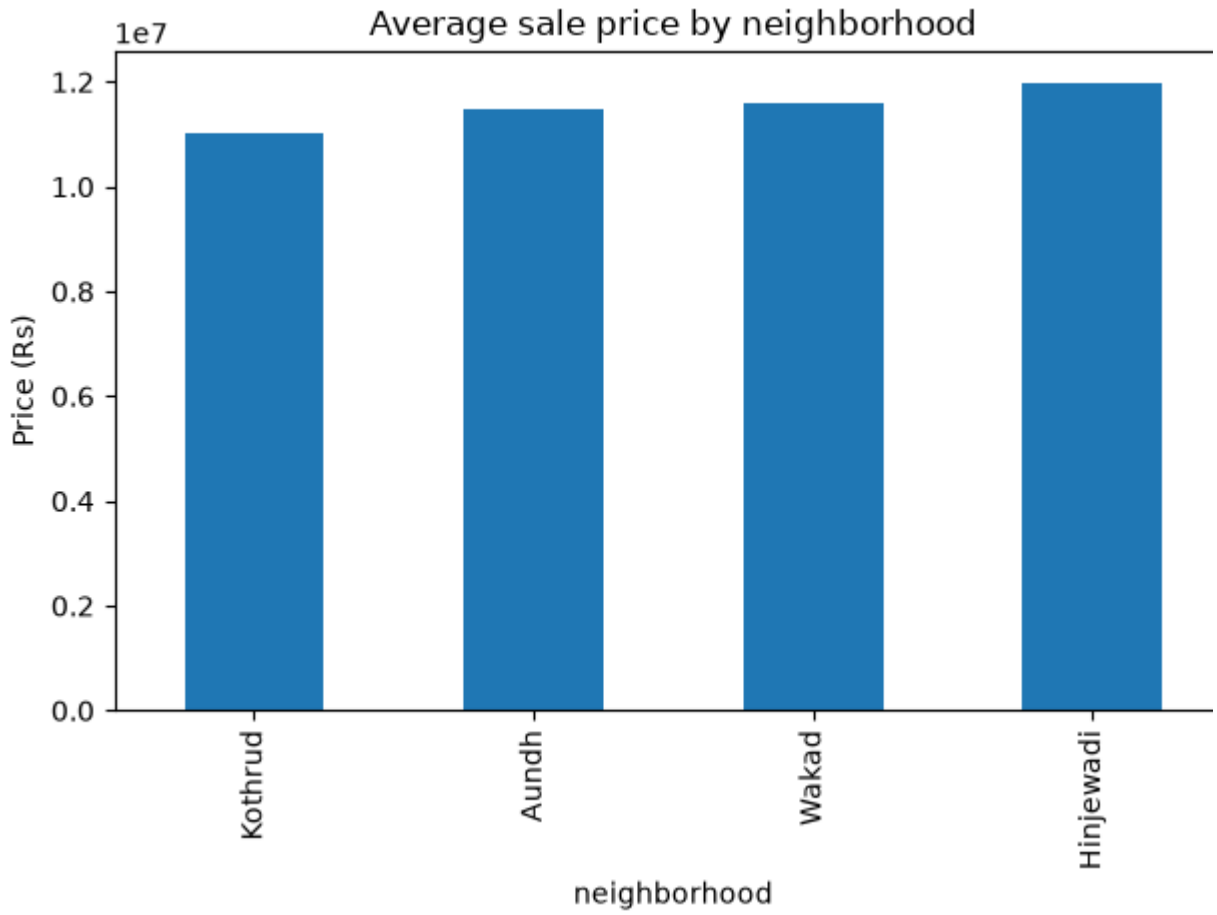
In [9]:
```
avg_nb = df.groupby("neighborhood")["sale_price_rs"].mean().sort_values()
print(avg_nb)
```

```
neighborhood
Kothrud      1.100382e+07
Aundh        1.147698e+07
Wakad        1.158093e+07
Hinjewadi    1.197643e+07
Name: sale_price_rs, dtype: float64
```

In [10]:
```
avg_type = df.groupby("property_type")["sale_price_rs"].mean()
print(avg_type)
```

```
property_type
Flat     1.225313e+07
Plot     1.136694e+07
Villa    1.088310e+07
Name: sale_price_rs, dtype: float64
```

In [11]:
```
avg_nb.plot(kind="bar")
plt.title("Average sale price by neighborhood")
plt.ylabel("Price (Rs)")
plt.tight_layout()
plt.show()
```




In [12]:
```
df = pd.get_dummies(df, columns=["neighborhood", "property_type"], drop_f
print("encoded shape:", df.shape)
```

```
encoded shape: (246, 10)
```

In [13]:
```
print("done")
```

```
done
```